

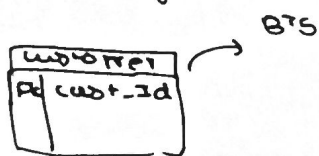
30 Sept

Porish

- ① the standard
- ② correct date-date
- ③ measure
- ④ sense the model
- ⑤ check

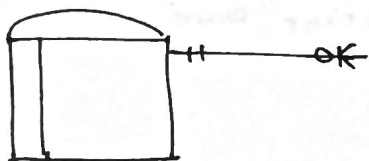
10 min

primary key ::



cust-ID	cust-name	...
1001		
1002		

attribute contains unique identifier of every record in the table



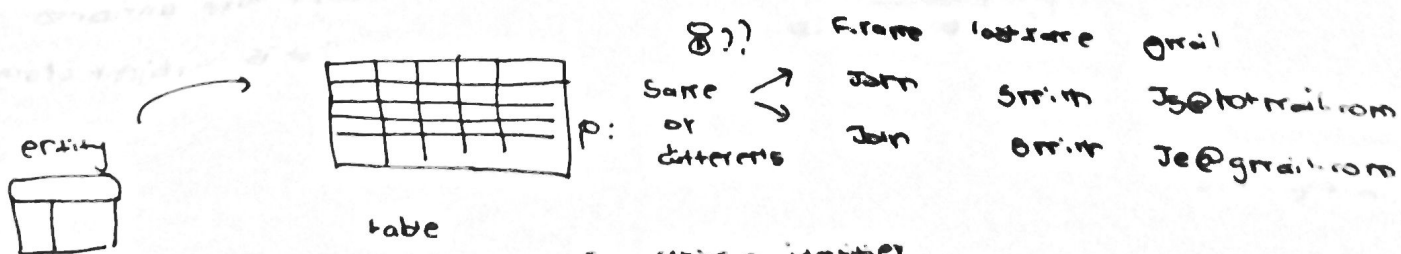
entity || attribute || relationship || cardinality

unique identifier distinguish each record from another

PK : Rules

- unique
- never change
- never null

Register - 1



primary key

- unique identifier
- never change (update)
- unique
- Not null

cust-ID is an attribute containing unique identifier

foreign key ::

- ↳ Same as primary key
- ↳ it just sits in foreign place

It helps us visually reinforce the fact

→ ~~1/2~~ F.K. Rules

- ① can be repeated
- ② Multiple F.K. in one entity

7:50 min

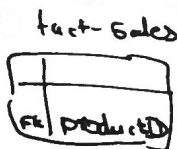
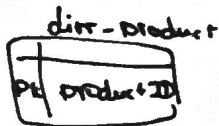
Revision - 1



P: Same or different
S: unique identifier
Primary key
Product ID

Foreign key:

when primary key of one table sits in foreign place;



composite

when two or more attribute are needed to create unique identifier for every record in your table

order-id product-id

PK	order ID
PK	product ID

Rule: fewer attributes

Don't take attributes

that is subject change

Bridge table

ERD? more specific

— Data model

Secure only what is need ;;

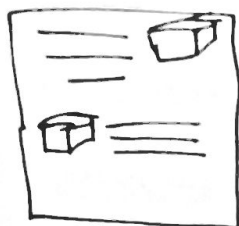
Don't force into every test

— 21 Sept

Day 1

Python

placeholder to store values that can be used later



variable make update easy
one place $\begin{cases} \rightarrow \text{affect multiple place} \end{cases}$

Pareto principle

→ where do the traffic accident happen?

[Different way representing frequency distribution to understand what is important]

Revision

Step 1: Reorganize the data

Step 2:

Reco

interpret &

recognize the

Pareto principle

— input() — inbuilt python function }

Python Data types

22/9/26 ①

Python needs to know how to put values in the box

a = 10
b = "Hello"

10
a

hello
b

memory

10
a

hello
b

→ Python automatically detect & data types

type: upper()

only accept:



a = 'Hello'

b

why Data types?

Revision - 1

different data type ::

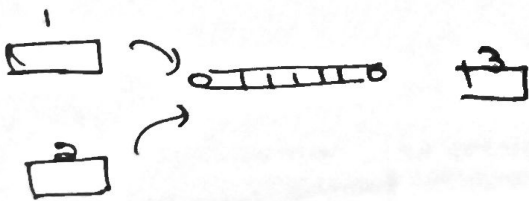
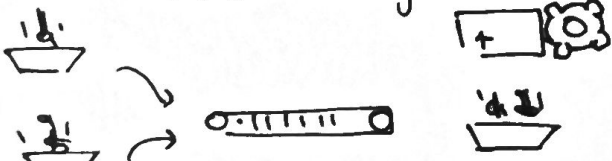


|| different data type ::



↳ concept belt problem (function)

↳ same conveyor belt will before differently



Type & Data types ::

No value

Single value

Multiple value

collections
container

" primitive data type

↳ list

↳ tuple

↳ dictionary

↳ set

[00]01

Sta



Analysing :

data

garbage

garbage

Type

• Des

• P

• O

lifecycle

(RISD - 20

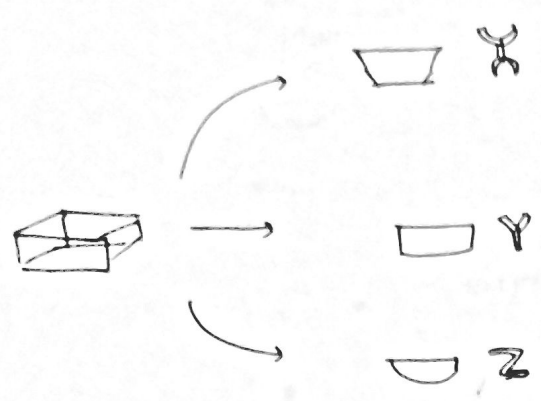
1) optin

2) Ser

3) P

But we have

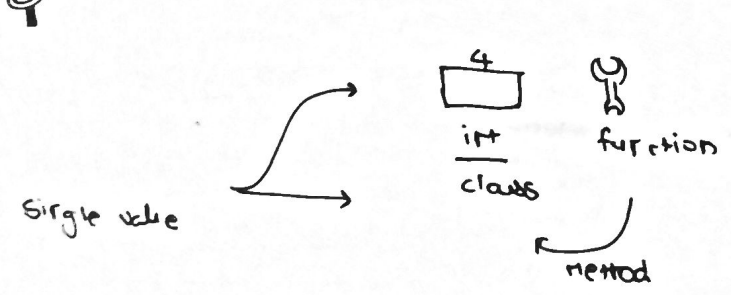
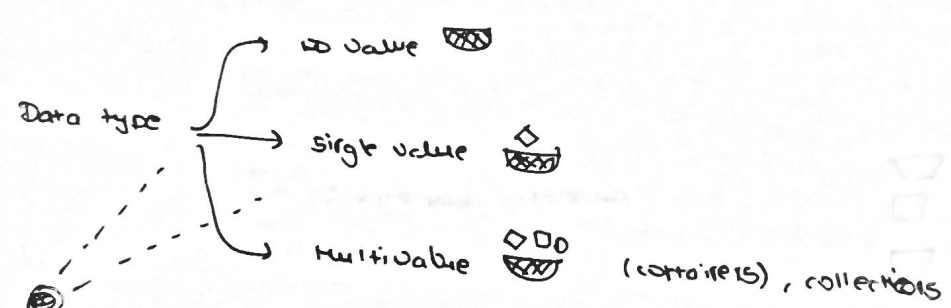
Do something to our data



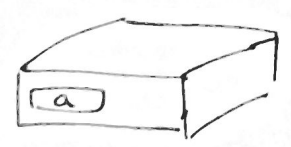
Data type

different box :: comes with it own set of tools

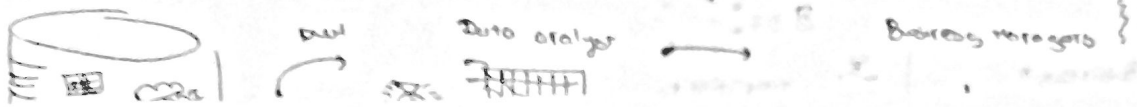
- x x -



Data type
different boxes :: & specific tool



Variable is a label



Analysis ::

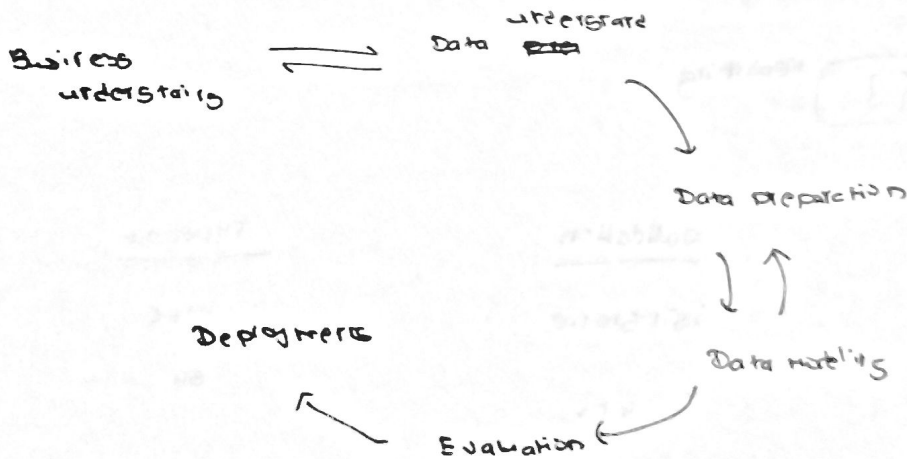


Type

- Descriptive
- predictive
- prescriptive



lifecycle



(CRISP - DM)

- 1) optimize revenue growth
- 2) Segment customers
- 3) pinpoint bottlenecks

Data we have



Data

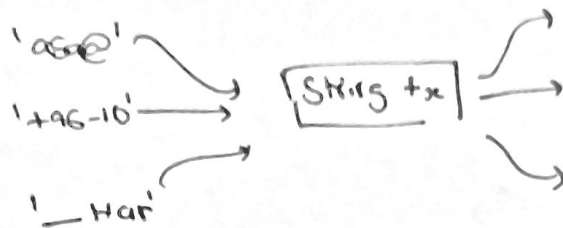
"Data Preparation"

python String (s)

DE

DA

DS



Text Handling

Manipulation

concat() or +

upper()

lower()

strip → lstrip
→ rstrip

replace

* 2 f { } New thing

Search

endswith

startswith

find()

in

① count -- frequency check
-- Query check

② Replace -- Numeric format

Calculation

len

count(, s, e)
optional

Extraction

index[]

index[: :]
s e
↑
Exclusive

String → 0-indexed

Validation

isinstance

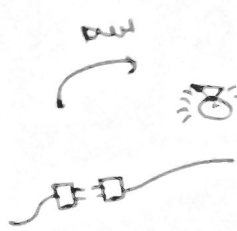
isalpha

isnumeric

Typecast

type

str



Excel

Business managers



was the business good?

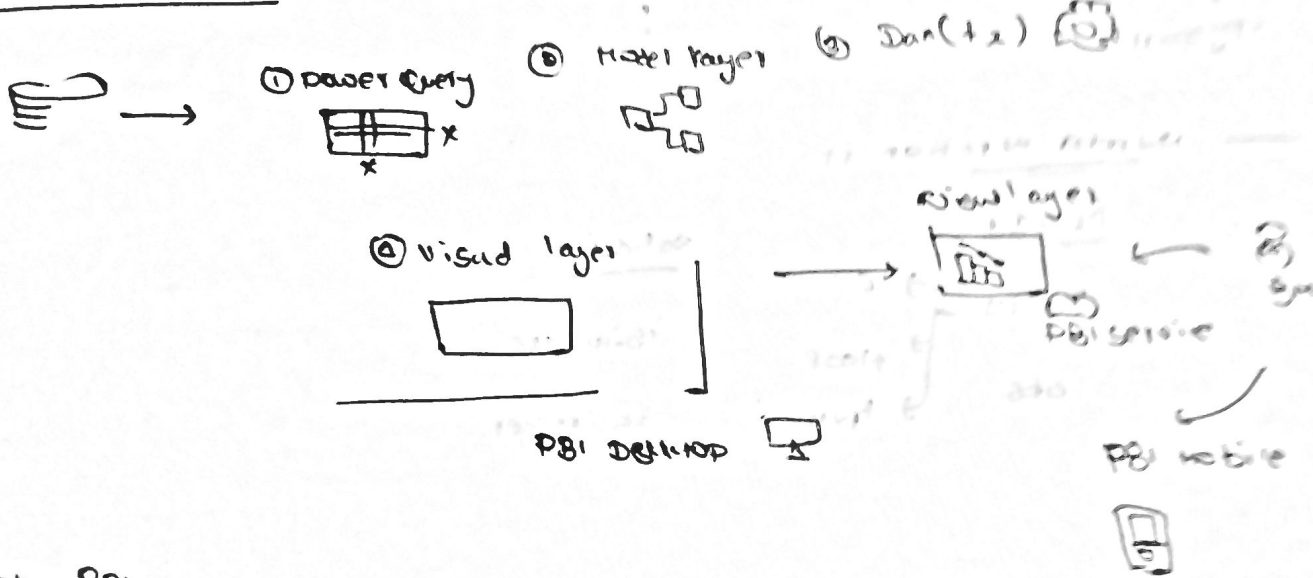
Problems:

- ↳ Manual
- ↳ outdated
- ↳ Hard to correct
- ↳ Hard to collaborate
- ↳ access (Security)

Solution BI

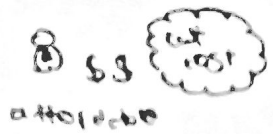
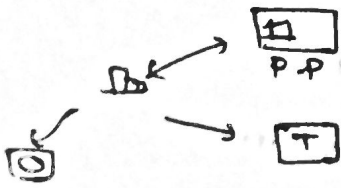
- ↳ automate
- ↳ Refresh
- ↳ easy to correct
- ↳ easy to collaborate (Security)
- ↳ access (RIG)

components of BI::



why PBIS

- ↳ correlated to your work
- ||
- ↳ easy to monitor



[PBIS] PBIS is a business intelligence tool

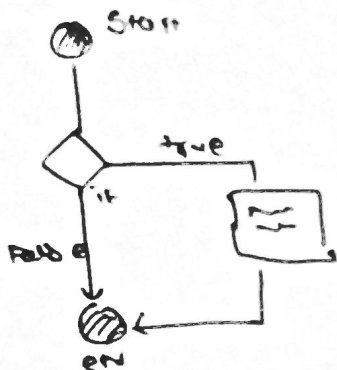
Standard report

Advanced report

conditional statements ::

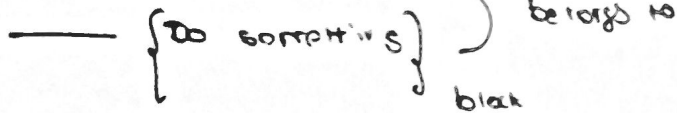
check point that creates a condition

① if statement it



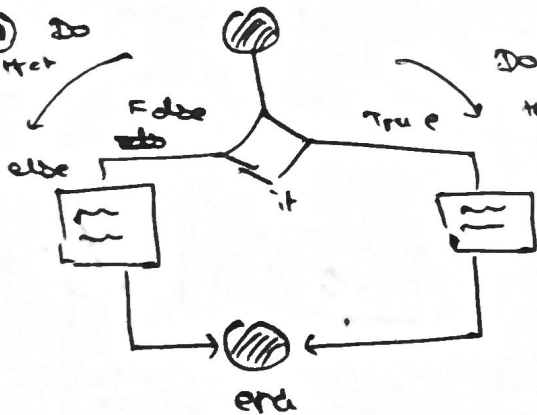
Indentation :: ::

if condition :



it's c.s, loop, variables

② if-else

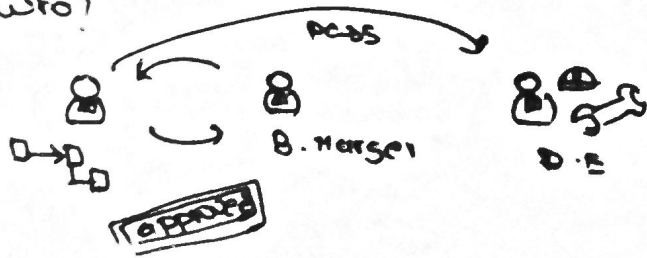


else runs only if all the previous condition is False

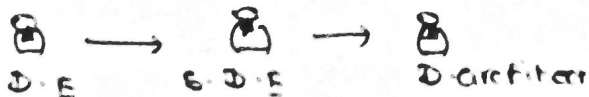
two way decisions

Data architecture

who?



current Road Map?



why visual?



vis tree test



more?



SS

A domain

□□□□ → Ore

SE

2319126

①

python -- string handling

Extraction → indexing

→ slicing

Search

SQL Problem Solving

case when

distinct

count

count

suspect

Marks

1

90

1

Null

50

Null

70

2

80

Null

90

2

90

2

3

70

Null

60

Null

50

Null

xxx

Q1::

1

✓

How Many Dates

How Many total

2

x

☐

☐

3

x

Q2::

☐

a

☐

b

☐

Each filter

Design Data architecture

①

24/9/36

Ass

task :

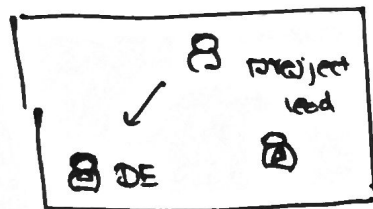
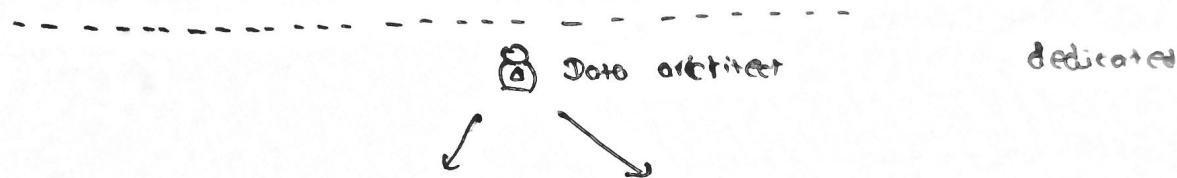
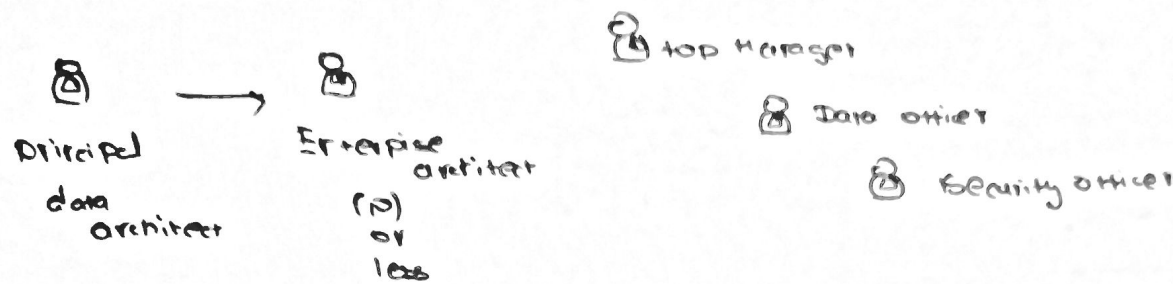
Date

Design data platform architecture

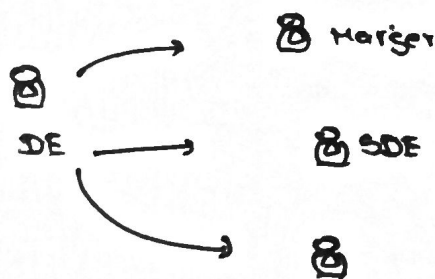
Data governance, catalog & cockpit

Design Data Model & Schema

7/15



Re



Do you know at your time if Google doing that

Vendor architecture ← platform & tools

Is this something you should provide solution to company

Can you understand the flow of data?

Data architect



Data used?



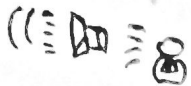
Export things usually

Text

Audio



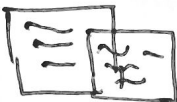
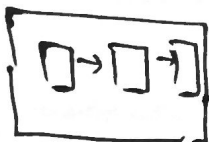
user



Engineer

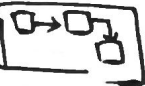
↳ imagination would be different

Visual are better way to communicate your plan to other than a BIG documents



Revision 1:

①



Explain the plan clearly

High level DP makes:

① Speaking Two language || [Problems → solution]

② New theory :: not Data only Projects

③ Adobe \$ \$?

④ E → [Diagram] → [Diagram]

Data mesh
Data lake

↑
clearly ideas

was not need?

⑤ Understand the business Needs :: 24 weeks

⑥ like are system [Diagram] work there it

VS #



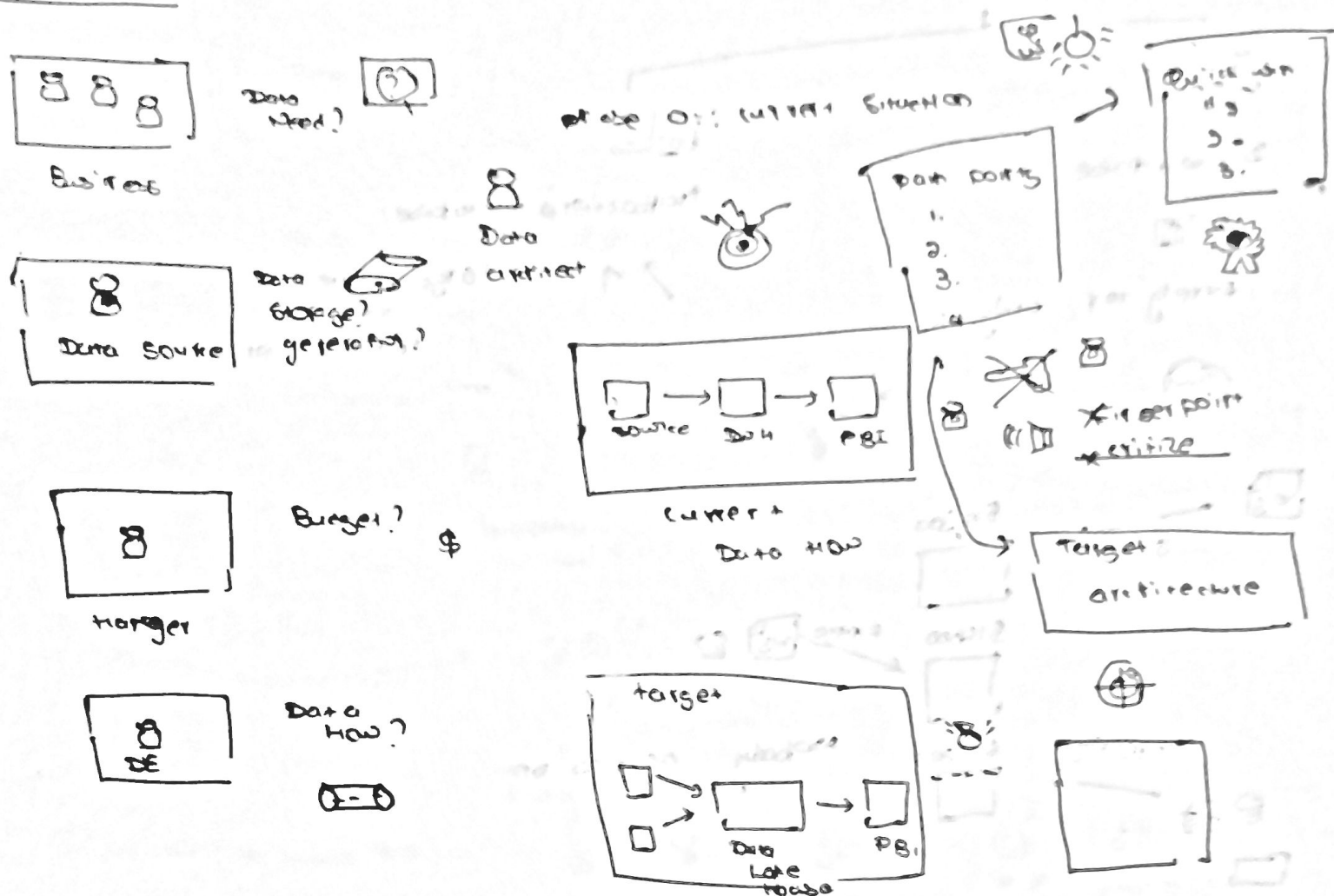
\$ \$

Q

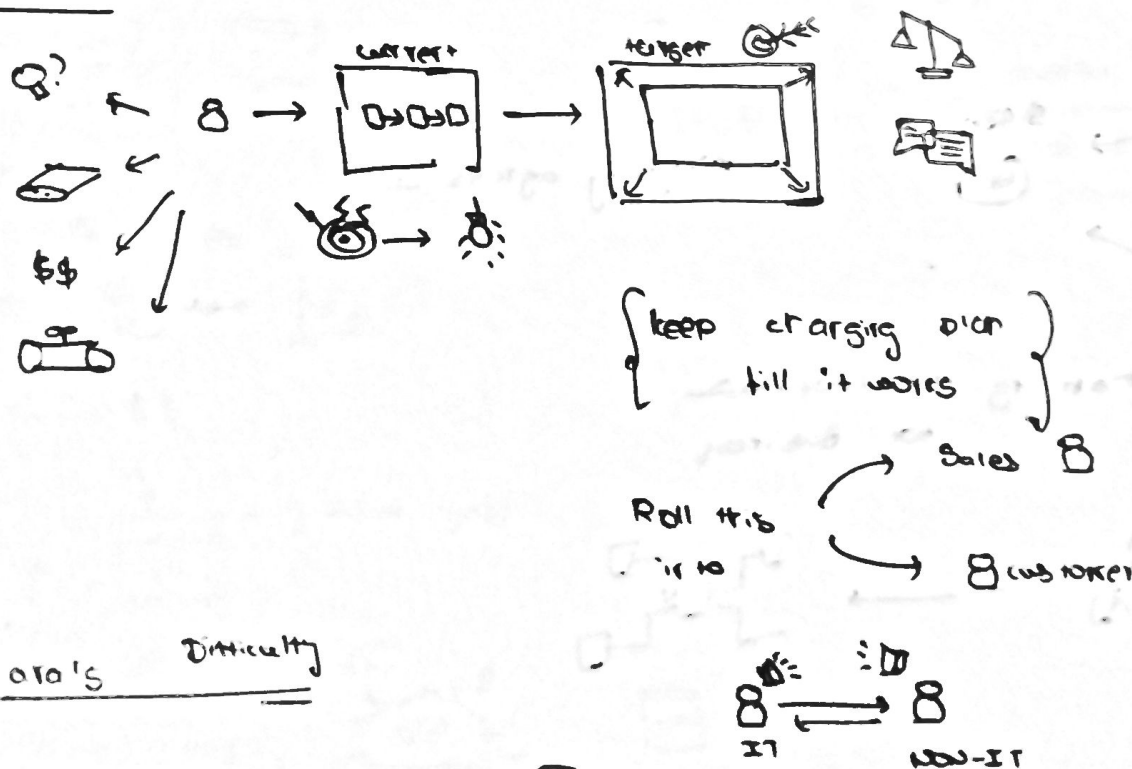
Send

25 Sept

Data architect



Review -



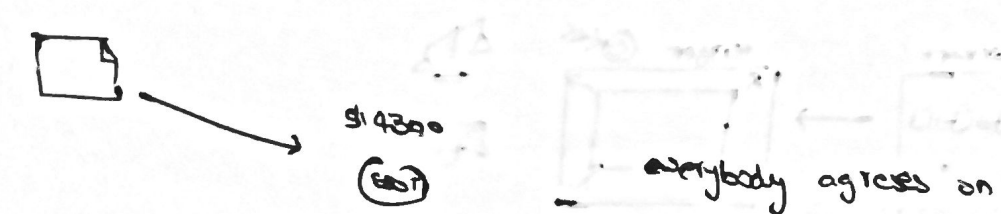
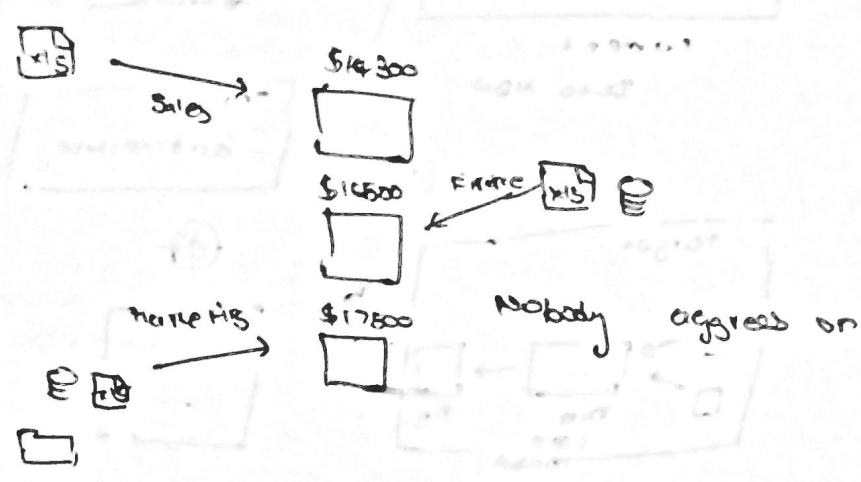
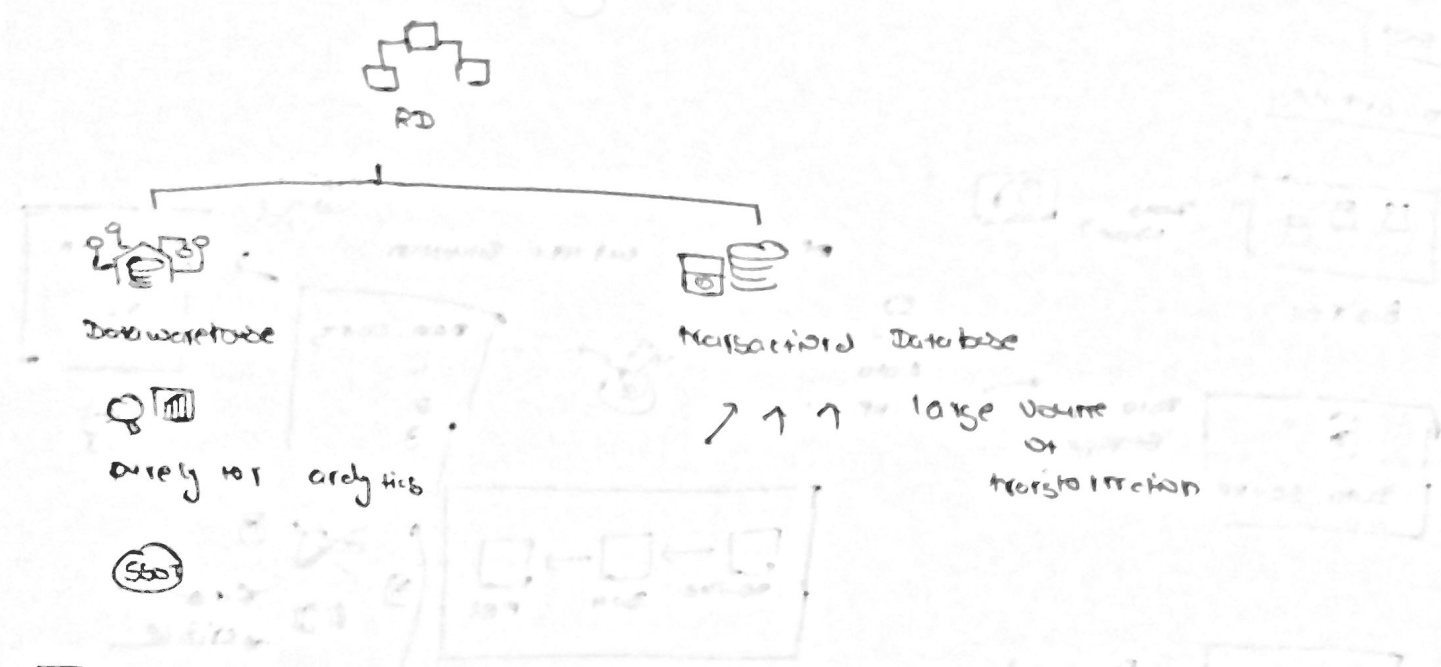
Baron's Difficulty

- ↳ Data model
- ↳ 3NF, Data Vault
- ↳ why Data Warehouse?

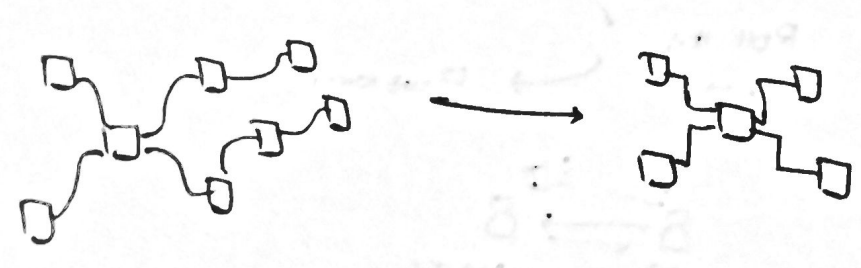


1:00

different
to communicate
documents
in the plan etc
usage 11 [prob
Data need
Data let
business Need
work



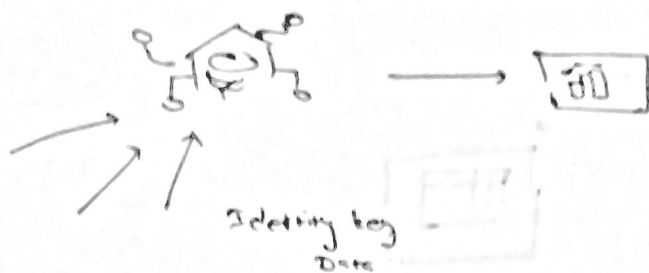
4) Simplicity
transferring ~~to developer~~
for Business



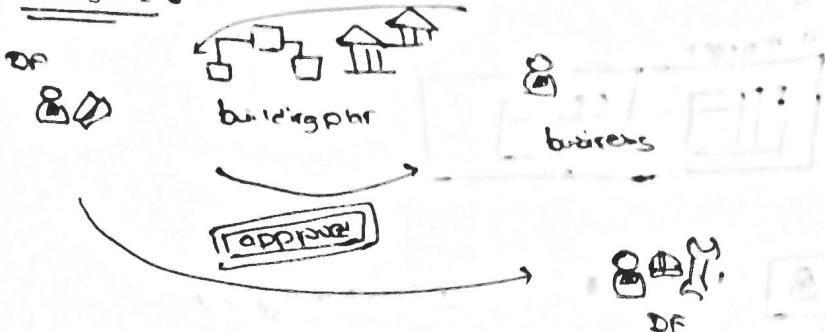
4) Data needs
update



25 Sept



Revisor - 2



cover :

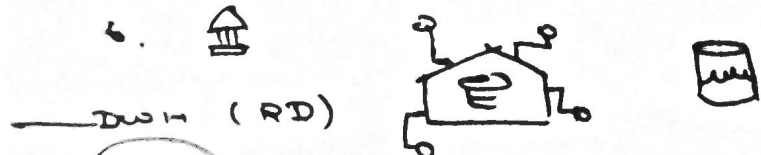
DE → SDE → DA

Draw :

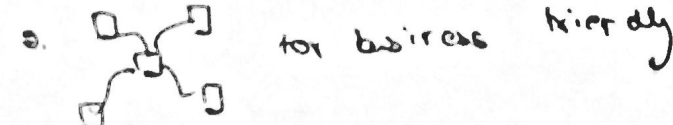


Migration :

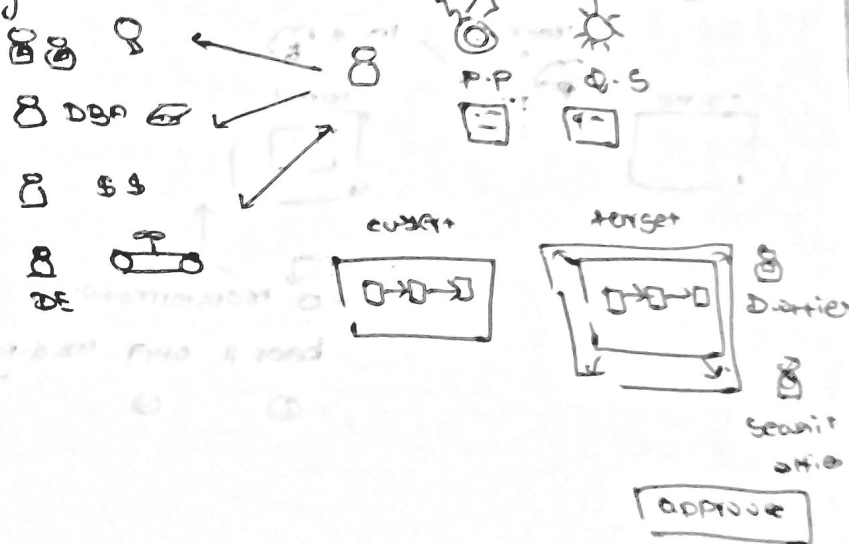
1. * & de
2. →
3. \$\$
4. A
5. SE } after job



1. SEOT



Today are

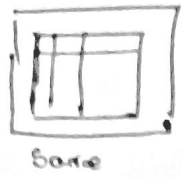
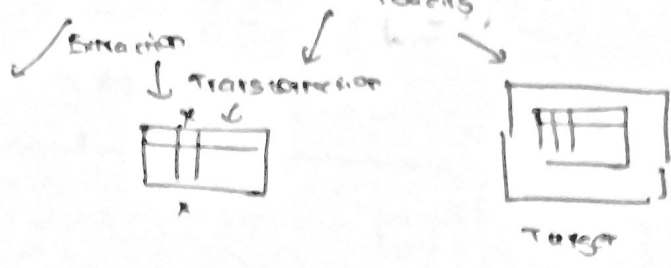


25 Sept

ent → asking

ET&ELT

Process

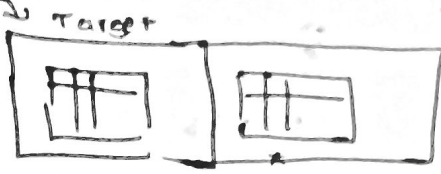
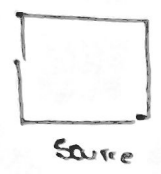


AI Engineer??

ELT

Extraction

Loading



AI Engineer

Data Analyst??

plan clearly

11 Problems

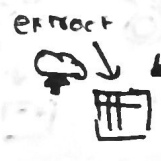
Data very proje

Data mesh
Data lake

1:54:26

Needs:

work data



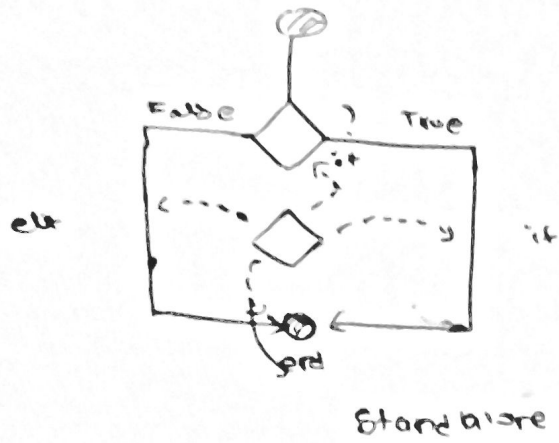
Transformation

before after loading

② ③

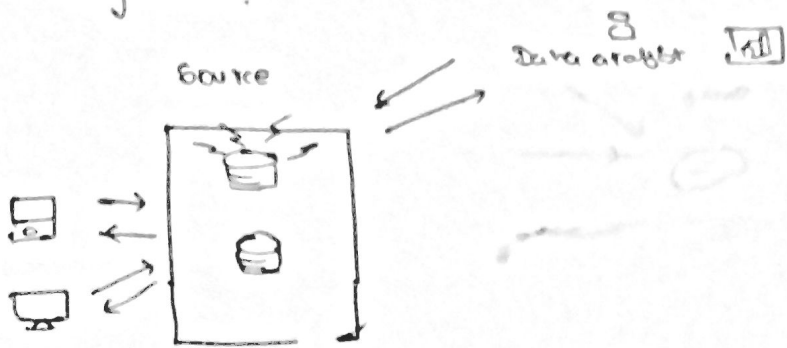
25 spot

ent $\rightarrow$ asking follow up question
again



why don't?

26/9/26

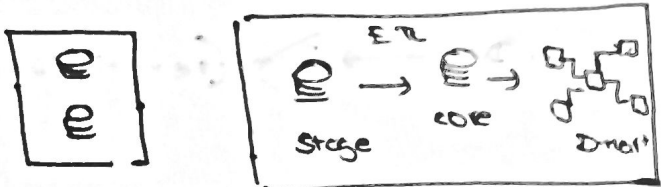


↳ Hoot

↳ no sample

↳ identity

Solution

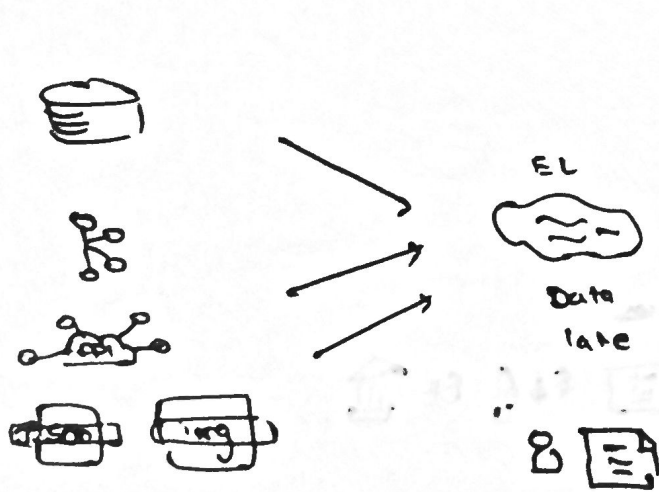


D.E. & D.Y.

DA & D.Y.

??
Data Scientist AI ??
Engineers

→→→ Datalake

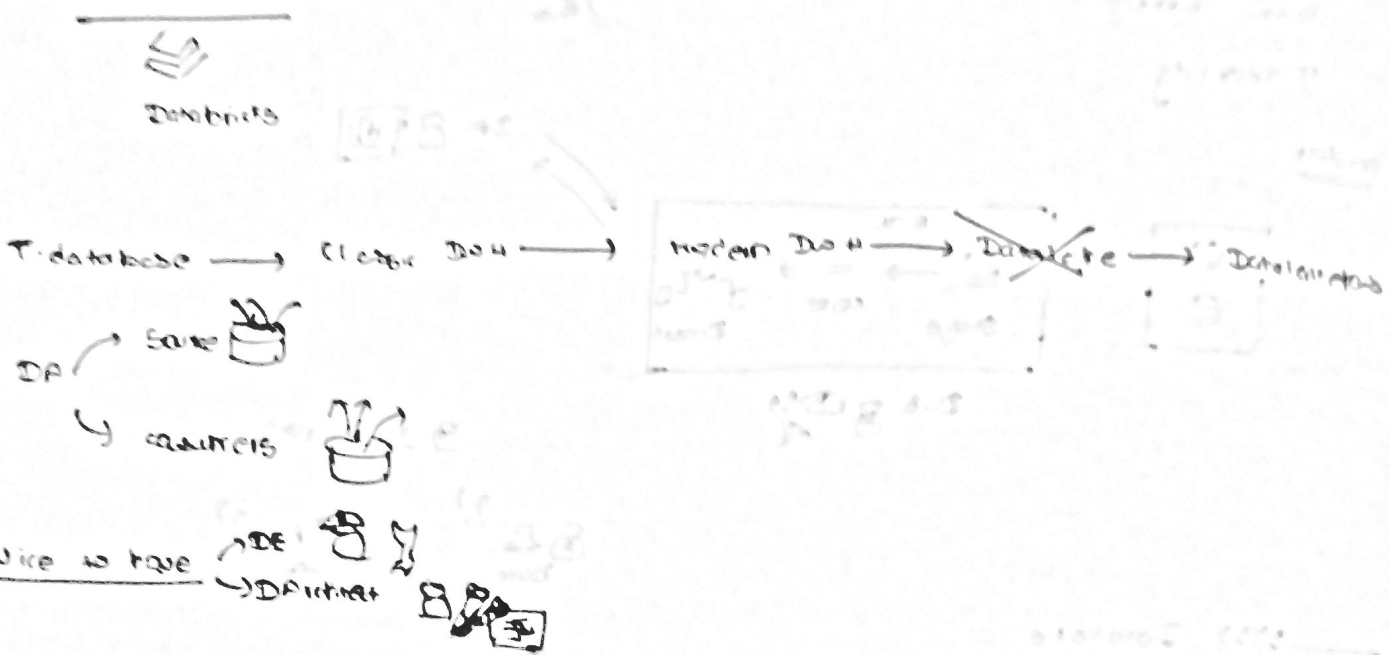
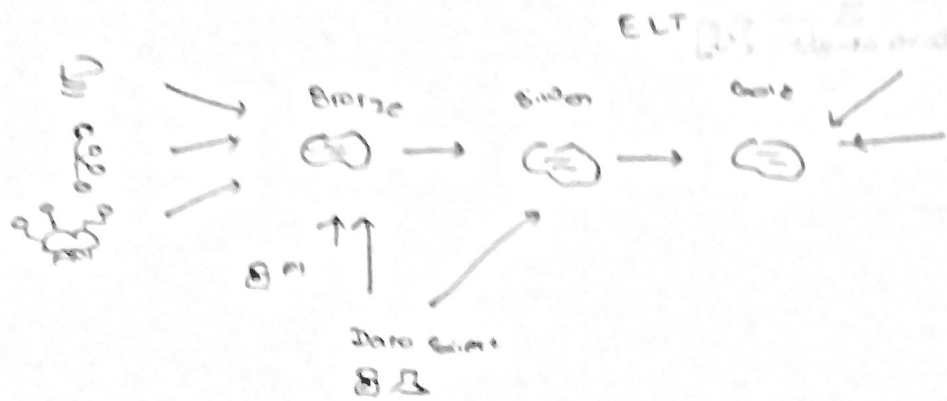


Transfer

↳ using things to do
↳ transform raw - extract

You don't care is

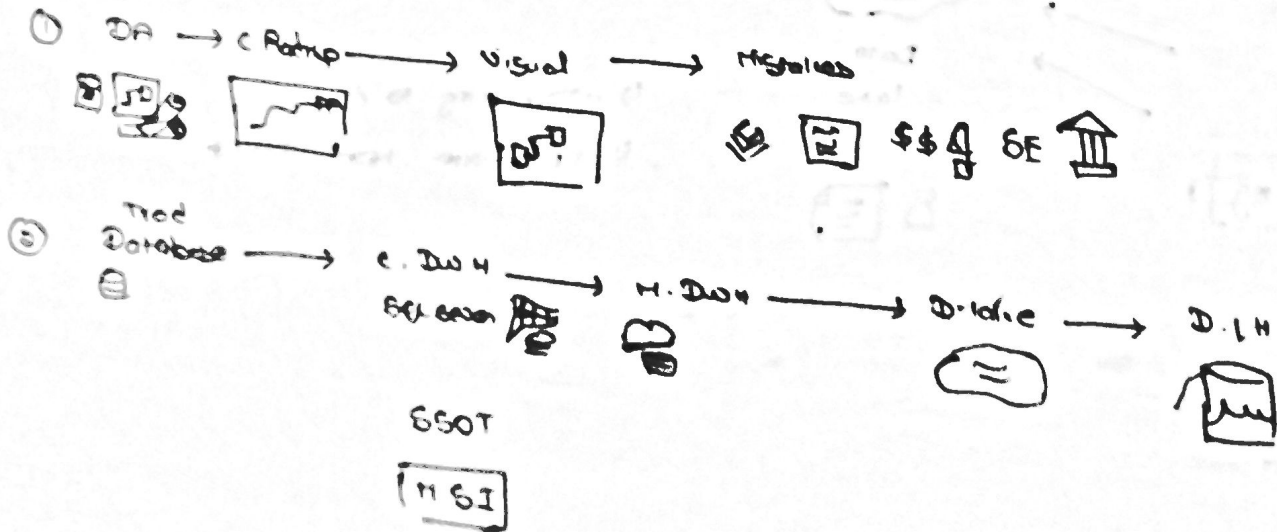
→→→ Datalake



xx ——— old days xx

a: 34, 53

Revisor - 2

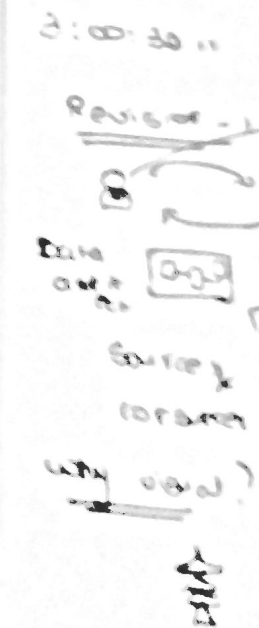
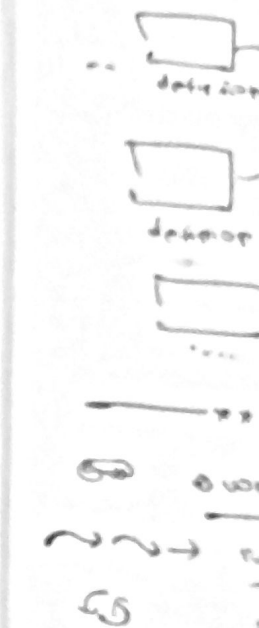


Data Product

understand

Prod

Is



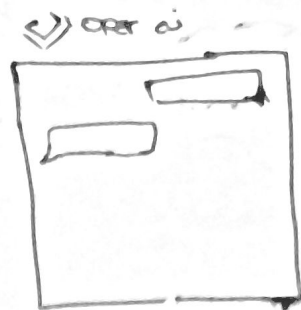
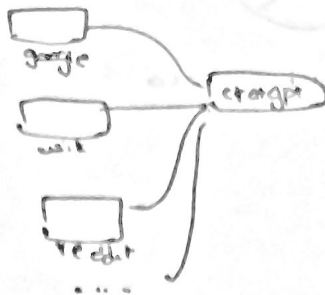
classical

Tradition

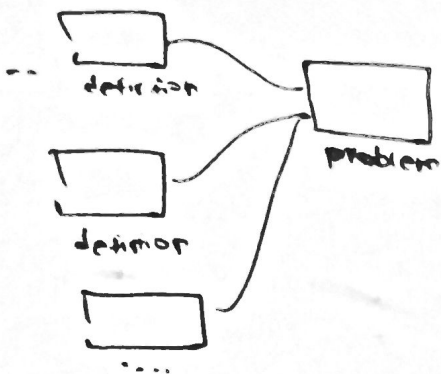
Data Product....

understanding the data product

- product?
- is it a data product

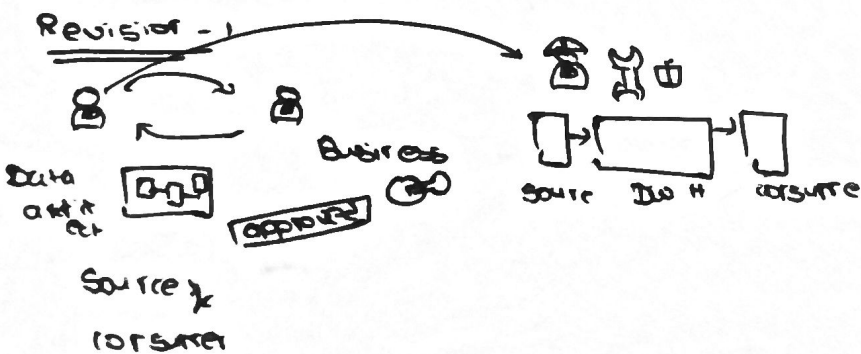


trust



	XX	XX	XX
	Ownership	Product	Reused
→	recycle		
→	Resubbing		

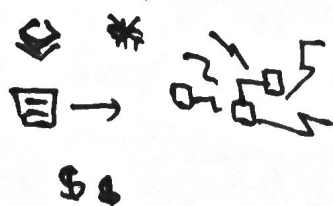
3:00:30



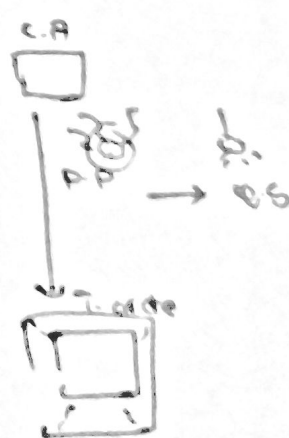
why view?



Migrates::



Day one



classical DWH → Modern DWH → Data Lake → Lake House



Tradition database

MSOT
Simplicity
Identity

Data Strategy



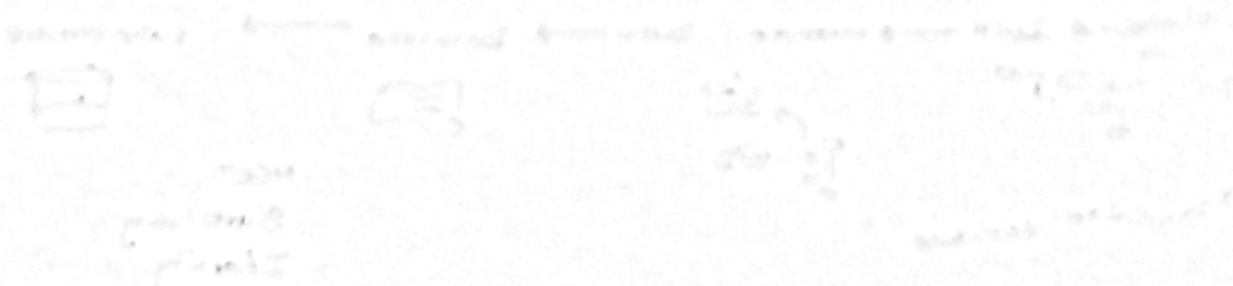
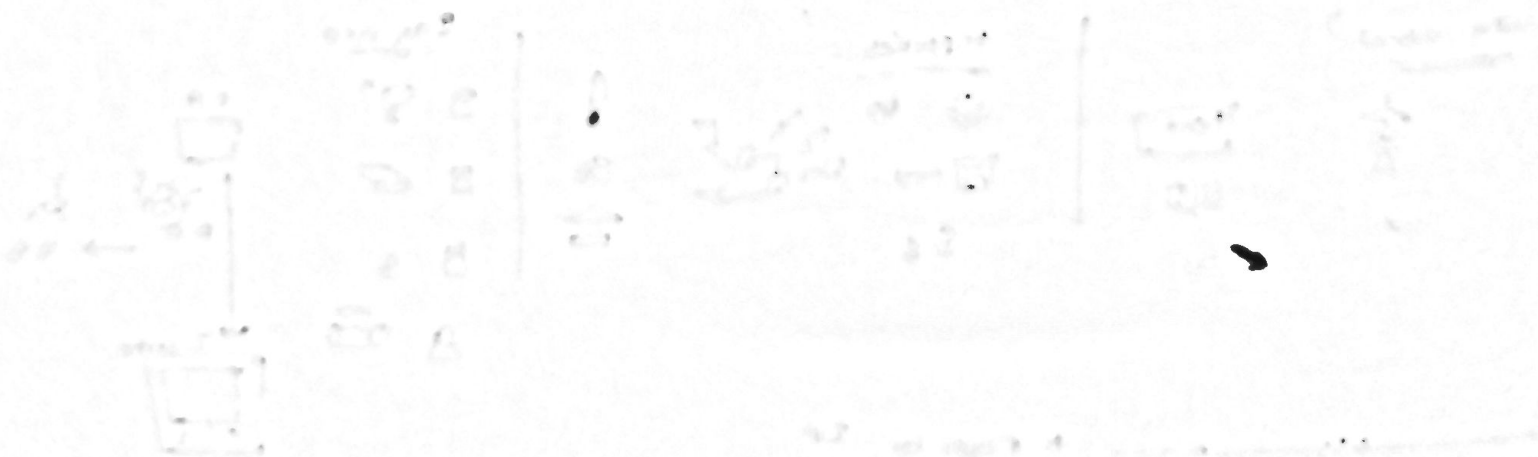
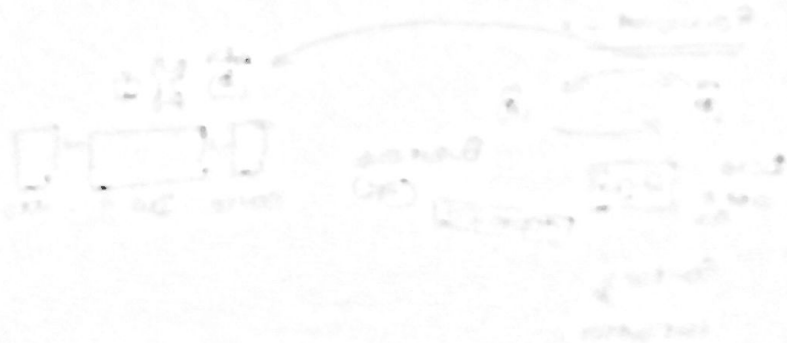
Hub & Spoke



Split by Domain
(Federated)



	1	2	3
1			
2			
3			



8 Data product owner



physical data



local catalog

Master Data

virtual

no tables

data catalog

Does about

what they enter



global catalog

Decentralized architecture (Data catalog)



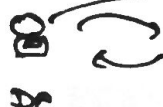
Data Platform Strategy



vision

3:21:40;;

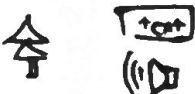
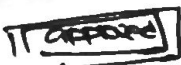
DA & its Migration



Business



DE



\$\$

Before plan



ISE

After plan

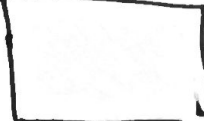


work too Dr

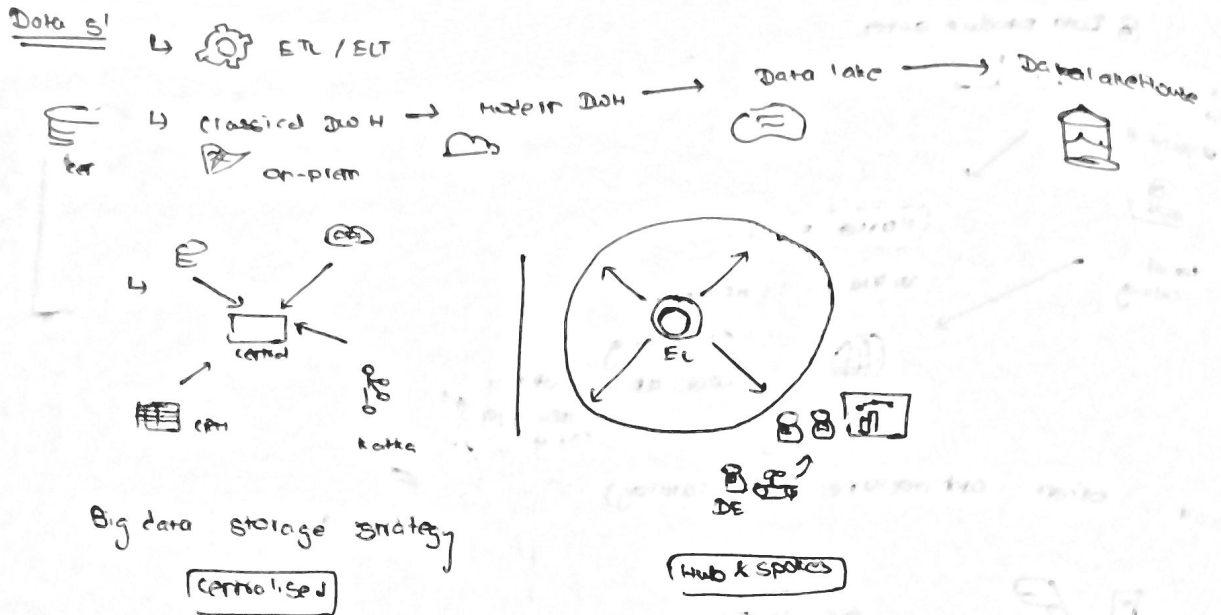


target plan

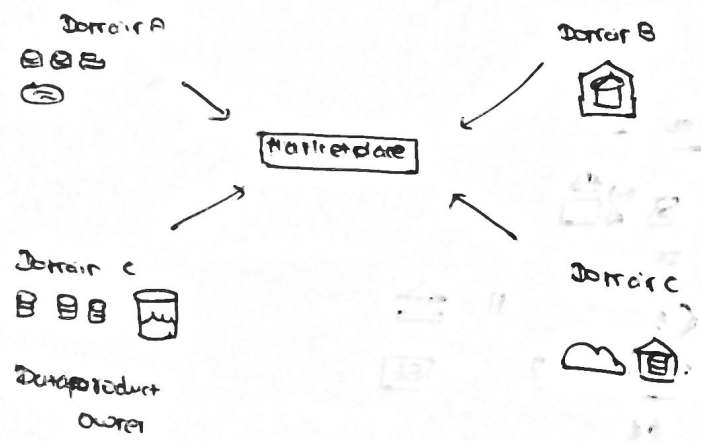
target



Data concepts



Big data storage strategy



Data mesh :: Data as a product

27 Sept ⑤

Data contract ::

Metadata :: Data

D. Product :: All

Customer

Decryption

Rows Size

Column Type

Describing how

Metadata

Dimensions

3:25 PM

